

PTS PTS90X3X SERIES INSTALLATION MANUAL

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PTS90X3X Series
PTS91X3X Series
PTS92X3X Series



120V/208V/240V AC Power Distribution Panel
60A - 125A Main Breaker
Suitable for Use as Service Equipment
Small Cell, Pico, Micro, CRAN or Macro
INSTALLATION GUIDE

Single SPD (L-G) per Leg or Dual SPD (L-N) per Leg, (N-PE)

PTS90X3X - 70A, 120V/208V/240V AC, 12 UL489 Breaker Positions

PTS91X3X - 100A, 120V/208V/240V AC, 16 UL489 Breaker Positions

PTS92X3x - 125A, 120V/208V/240V AC, 12, 16 or 20 UL489 Breaker Positions





PTS9XX3X Series Small Cell 120/240V AC Power Distribution Load Centers Installation Guide

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1.0 Model Overview

The PTS9XX3X Series of Small Cell 120/240V AC Power Distribution Panels with optional External Disconnect Switch, provides a custom factory fully assembled load center for Small Cell, Pico, Micro, CRAN and Macro Deployments.

The PTS9XX3X Series products are SUSE Rated:
Suitable for Use as Service Equipment and conform to UL standards.



PTS9XX3X Series Small Cell AC Power Distribution Load Centers





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PTS9XX3X – Example of Dimensional Drawings

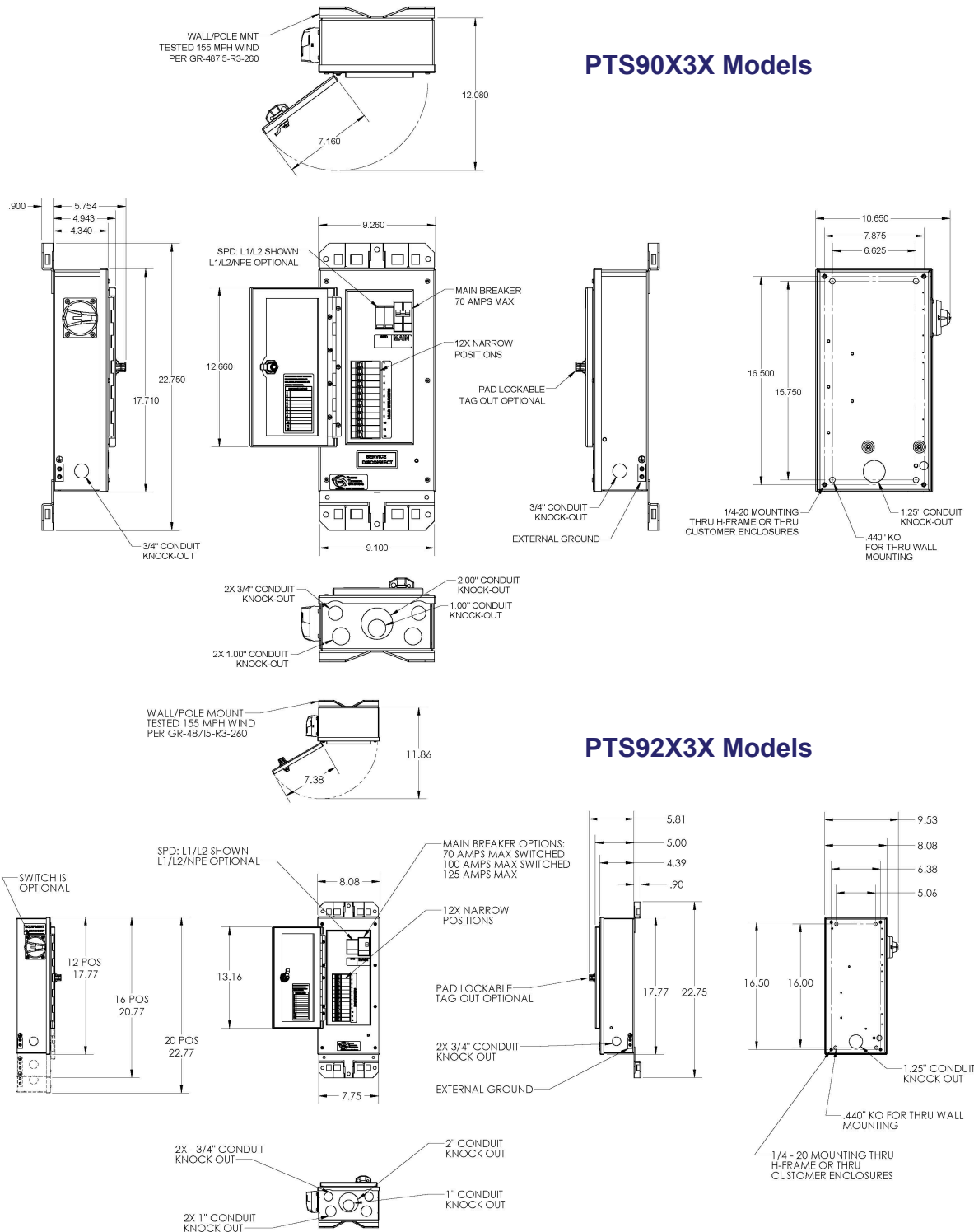


Figure 1: Example Dimensional Drawings





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1.1 Specifications

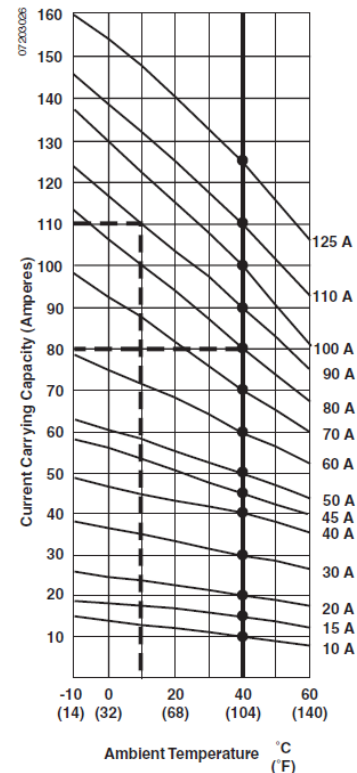
Electrical Specifications:

Panel Rating	120/208/240V AC
Phase	Single Phase
Load Center Breakers	(12, 14, 16, 20 Pos.) UL489
Load Center Input	2/0 – 14 awg, model specific
Breaker Connections	4 awg – 14 awg
Approvals	UL67, Type 3R

Mechanical Specifications:

Color	Grey Standard
Size(in) 12 - 14pos	17.71 × 9.26 × 4.94
Size(in) 16 pos	20.71 × 9.26 × 4.94
Size(in) 20 pos	23.71 × 9.26 × 4.94
Weight(lbs)	12 to 16lbs
Temperature(°C)	-40 to +40, above +40 see chart
Operating Humidity	5~95%

Ambient Derating of QOU Circuit Breaker



Caution: Installation shall be completed per all NEC and Local Codes. Do not install PTS9000 Systems before reading and understanding all specifications and installation guides provided. Installation should be performed by a qualified installer.





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Breaker Load Out Examples:

		Position												M	Description			
Base P/N		P1	P2	P3	P4	P5	P6	P7	P8	P9	P10	P11	P12					
PTS90X36	- 6 2 1	-	7	7	7	7								- P2	240VAC / 2 Pole - with Integrated External Disconnect Switch			
PTS90X36	- 7 2 1	-	G	G	G	7								- P2	240VAC / 2 Pole - with Integrated External Disconnect Switch			
PTS90X36	- 7 2 1	-	Q		Q		Q	7	7					- P2N	240VAC / 2 Pole - with Integrated External Disconnect Switch			
PTS90X39	- 4 2 1	-	7	7	7	7								- P2	240VAC / 2 Pole - No Integrated External Disconnect Switch			
PTS90X39	- 6 2 1	-	G	G	G	7								- P2	240VAC / 2 Pole - No Integrated External Disconnect Switch			
PTS90X39	- 7 2 1	-	Q		Q		Q	7	7					- P2N	240VAC / 2 Pole - No Integrated External Disconnect Switch			
PTS90X39	- 7 2 1	-	V		V	K	K	K	K	K	K	K	K	- P2N	240VAC / 2 Pole - No Integrated External Disconnect Switch			
P2 = SPD (L1-N/PE and L2-N/PE) P2N = Dual SPD (L1-N, L2-N and N-PE)																		
		1Pole	1	2	3	4	5	6	7	8	9	A	C	F	G	K	S	Open
		2Pole					W					L	N	Q	R	V	T	
		AMP	1	2	3	4	5	6	7	8	9	10	12	15	16	20	25	Blank
		= Phase (1=Single Phase)																
		= Poles (1=120VAC, 2=240VAC)																
		= Amperage Code (3=30A, 4=40A, 5=50A, 6=60A, 7=70A) 10 kA Main Breaker (22kAIC Optional)																

Position																										
Base P/N		P1	P2	P3	P4	P5	P6	P7	P8	P10	P12	P16	P20	-	M	Description										
PTS92X36	- 6 2	-	7	7	7	7								-	P2	240V/60A 2 Pole - with Integrated External Disconnect Switch										
PTS92X36	- 7 2	-	G	G	G	7								-	P2	240V/70A 2 Pole - with Integrated External Disconnect Switch										
PTS92X36	- 8 6	-		Q		Q		Q	7	7				-	P2N	240V/80A 2 Pole - with Integrated External Disconnect Switch										
PTS92X36	0 0	-	T		T		T	7	7					-	P2N	240V/100A 2 Pole - with Integrated External Disconnect Switch										
PTS92X39	- 7 2	-	7	7	7	7								-	P2	240V/70A 2 Pole - No External Switch										
PTS92X39	- 7 2	-	G	G	G	7								-	P2	240V/70A 2 Pole - No External Switch										
PTS92X39	- 8 2	-		Q		Q		Q	7	7				-	P2N	240V/80A 2 Pole - No External Switch										
PTS92X39	- 0 6	-		Q		Q		Q	7	7				-	P2N	240V/100A 2 Pole - No External Switch										
PTS92X39	- 2 0	-	V		V	V	A	A	A	A	7	7		-	P2N	240V/125A 2 Pole - No External Switch										
P2 = SPD (L1-N/PE and L2-N/PE) P2N = Dual SPD (L1-N, L2-N and N-PE)																										
		1Pole 1 2 3 4 5 6 7 8 9 A C F G K S														Open										
		2Pole 1 2 3 4 5 6 7 8 9 10 12 15 16 20 25														Blank										
		AMP 1 2 3 4 5 6 7 8 9 10 12 15 16 20 25																								
= Load Breaker Positions (2=12pos, 6=16pos, 0=20pos)																										
= Amperage Code (6=6-A, 7=70A, 8=80A, 0=100A, 2=125A) 10 kA Main Breaker, 60A/22kA Optional																										

Additional/Spare Breakers		
PTS P/N	Description	
PT-KM1N-07A	UL489A DIN Rail Breaker, 07 AMP	
PT-KM1N-10A	UL489A DIN Rail Breaker, 10 AMP	
PT-KM1N-12A	UL489A DIN Rail Breaker, 12 AMP	
PT-KM1N-16A	UL489A DIN Rail Breaker, 16 AMP	
PT-KM1N-20A	UL489A DIN Rail Breaker, 20 AMP	
PT-KM1N-25A	UL489A DIN Rail Breaker, 25 AMP	
PT-KM2N-05A	UL489A DIN Rail Breaker-Dual-Pole, 05 AMP	
PT-KM2N-10A	UL489A DIN Rail Breaker-Dual-Pole, 10 AMP	
PT-KM2N-15A	UL489A DIN Rail Breaker-Dual-Pole, 15 AMP	
PT-KM2N-20A	UL489A DIN Rail Breaker-Dual-Pole, 20 AMP	
PT-KM2N-25A	UL489A DIN Rail Breaker-Dual-Pole, 25 AMP	



Caution: Installation shall be completed per all NEC and Local Codes. Do not install PTS9000 Systems before reading and understanding all specifications and installation guides provided. Installation should be performed by a qualified installer.





PTS9XX3X Series Small Cell 120/240V AC Power Distribution Load Centers Installation Guide

1.3 Application

The PTS9XX3X are compact 120V/208V/240V AC Load Centers designed for Small Cell, Pico, Micro, CRAN and Macro Deployments where distribution of AC power is required.

The PTS9XX3X Series of compact AC Load Centers can also be used for AC Power Distribution applications where limited space exists, excluding the ability to use a larger load center.

The PTS9XX3X Series are Suitable for Use as Service Equipment.

The PTS9XX3X Series of compact AC Load Centers can also be used as a sub-tending branch panel when modernizing or expanding existing sites.

The PTS9XX39 Series options include an external rotary disconnect switch, allowing for the utility, fire safety and first responders to quickly and safely disconnect power.

2.0 Features

The PTS9XX3X Series of Small Cell AC Load Centers support a customer specific configuration, ready for install in new or expansion environments.

2.1 Main Breaker

The PTS9XX3X Systems can support a QOU main breaker with Ampacity from 70A to 125Amps, either in Single Pole 120VAC, or Dual Pole 208V/240VAC.

The main breaker temperature de-rating chart is found on page 6

2.2 Distribution Breakers

The PTS90X3X Systems have 12 UL489 Breaker positions

The PTS91X3X Systems have 16 UL489 Breaker positions

The PTS92X3X Systems have 12, 16 or 20 UL489 Breaker positions

2.3 Optional SPD Protection

Single SPD (L1-N and L2-N) or a Dual SPD (L1-N, L2-N and N-PE)

2.4 Optional External Disconnect Switch

The PTS9XX36 Series does not include the external rotary disconnect switch and can support up to 125 Amps

The PTS9XX39 Series includes an external rotary disconnect switch and can support up to 100 Amps.





PTS9XX3X Series Small Cell 120/240V AC Power Distribution Load Centers Installation Guide

3.0 Inspecting and Unpacking

- 3.0.1 Inspect the packaging for obvious signs of rough handling and/or external damage.
- 3.0.2 Review Installation Guide.
- 3.0.3 Remove from plastic packaging.

4.0 Installation

The PTS9XX3X Systems are certified for outdoor installations.

This section provides installations guidelines to insure the appropriate requirements are met.

- 4.0.1 To maintain 3R Rating, the enclosure must be mounted vertically,
- 4.0.2 Do not make holes in top or sides where internal parts restrict access.
- 4.0.3 Use only raintight connectors
- 4.0.4 Panelboard intended to be field installed on a utility pole, street pole, shelter or flat surface suitable for installation per building code where applicable.

4.1 Mounting Bracket

- 4.1.1 A mounting bracket is included and pre-installed along the top and bottom back edge of the enclosure and can be used as a template, or mounting detail from specifications page can be used if preferred.
- 4.1.2 If mounting bracket is replaced, a lock-tite like adhesive should be applied to the screw threads and tighten according to specifications.

4.2 Mount enclosure vertically on surfaces

- 4.2.1 To maintain 3R Rating, the enclosure must be mounted vertically, do not make holes in top or sides, and use only raintight connectors
- 4.2.2 Customer is responsible to identify suitable location for mounting.
- 4.2.3 Mark and drill anchors or clips to secure enclosure to surface or pole
- 4.2.4 Torque to mounting hardware specifications.

4.3 Removing Front Door / Access Cover

- 4.3.1 Locate Front Door / Access Cover set screws and remove
- 4.3.2 Set aside for reuse.
- 4.3.3 Review Equipment Ground Connection Points
- 4.3.4 Review Neutral Connection Points





PTS9XX3X 120/240V AC Power Distribution Load Centers Installation Guide

4.4 Wiring Ground Connections

- 4.4.1 Ensure enclosure is grounded as per NEC and Local Jurisdictions Code requirements.
- 4.4.2 Ensure that the neutral buss bar is grounded per NEC and Local Jurisdictions Code if required.
- 4.4.3 Tighten connections as per manufacturers specifications.



Figure 3: Location of External Enclosure Ground Connection
Lower Left or Right Side on PTS9000 Series Enclosures



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PTS9XX3X Series Small Cell 120/240V AC Power Distribution Load Centers Installation Guide

4.5 Wiring Service and Load Connections

4.5.1 Wiring to be done in accordance with NEC and Local Jurisdictions Code.

4.5.1.1 Field conductors should be 70C copper minimum.

4.5.1.2 See QOU Breaker derating chart for proper operating ampacity on page 6.

4.5.2 Wire the Service Ground to the internal ground bar

4.5.3 Wire the Service Neutral to Neutral bar connection according to use and Local Jurisdictional Code.

4.5.4 If wiring for Service Entrance, ensure that the provided bond wiring strap is connected between the internal ground bar and neutral bar, and label accordingly as per 4.5.11.1 below.

4.5.5 Wire the Service Feed to the bottom side (service side) of the Main Breaker (s)

4.5.6 Land Equipment ground to ground bar

4.5.7 Land Equipment Neutral to Neutral bar

4.5.8 Land Load feeds to the right side (load side) of load breaker(s)

4.5.9 Ensure connections are properly seated in set screw capture ring openings prior to tightening per specifications.

4.5.10 Tighten per breaker and bar specifications

4.5.11 Ensure all breakers are labeled

4.5.11.1 If installed as Service Equipment, apply the "SERVICE DISCONNECT" label at the primary breaker positions

4.5.11.2 If installed as Branch Equipment, apply the "MAIN DISCONNECT" label at the primary breaker positions

4.5.12 Label Load Breakers accordingly

4.5.13 If completed, re-install the front cover.

4.5.14 Tighten screws to 17 - 18 inch pounds



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PTS9XX3X Series Small Cell 120/240V AC Power Distribution Load Centers Installation Guide

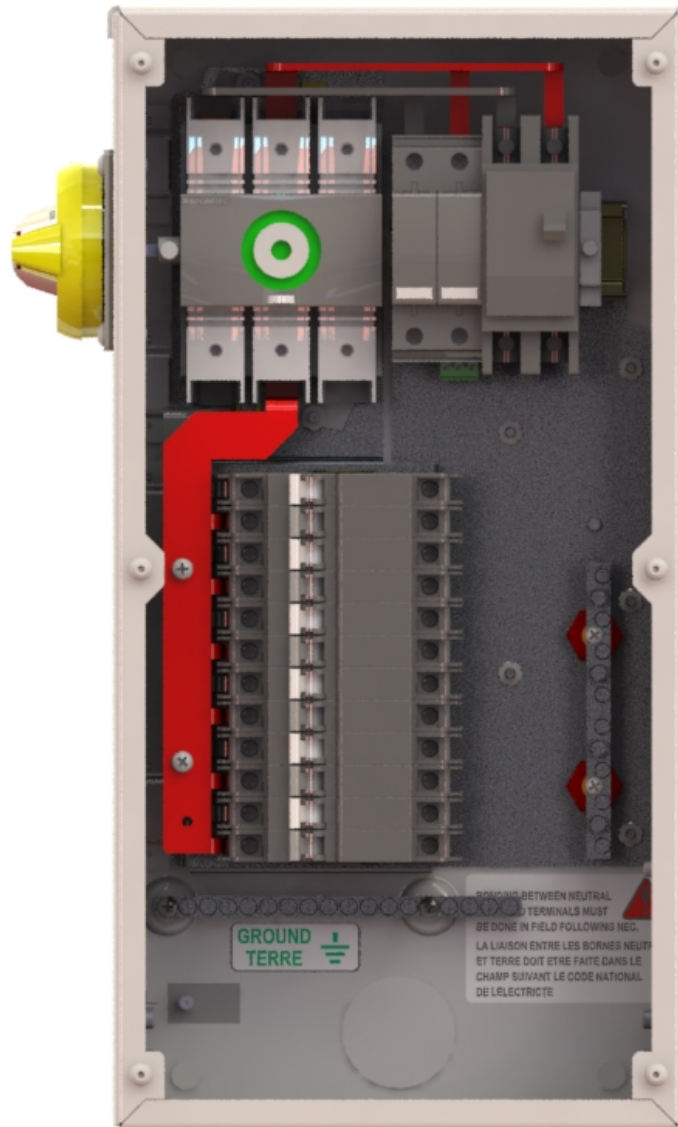


Figure 4: Standard Wiring Space Example



Caution: Installation shall be completed per all NEC and Local Codes. Do not install PTS90000 Systems before reading and understanding all specifications and installation guides provided. Installation should be performed by a qualified installer.





PTS9XX3X Series Small Cell 120/240V AC Power Distribution Load Centers Installation Guide



**Figure 5: PTS9XX39 Series External Disconnect Switch
Location (Yellow/Red Rotary Switch)
with Tag Out / Lock Out (White Tab) Function**



Caution: Installation shall be completed per all NEC and Local Codes. Do not install PTS9000 Systems before reading and understanding all specifications and installation guides provided. Installation should be performed by a qualified installer.





PTS9XX3X Series Small Cell 120/240V AC Power Distribution Load Centers Installation Guide

5.0 Installation Review

Inspect the installation, verifying that all is in accordance with this document, NEC, and local Codes.



Caution: Installation shall be completed per all NEC and Local Codes. Do not install PTS9000 Systems before reading and understanding all specifications and installation guides provided. Installation should be performed by a qualified installer.





PTS9XX3X Series Small Cell 120/240V AC Power Distribution Load Centers Installation Guide

6.0 Document and Revision Control

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